Rolling Admissions MDP

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1 Setup and Assumptions

Discrete time, finite horizon ($t = 1, 2, \dots, T$).

Each applicant belongs to one of $r \in N$ quality tiers. A student in tier $i \in \{1, 2, \dots, r\}$ has a discretized quality value of $q_i = \frac{i}{r}$. If a student with quality q_i matriculates, the school receives a terminal value proportional to q_i.

If a student applies, she is immediately placed onto a waitlist. The waitlist at time t is described by the vector $\mathbf{n}_t = (n_{1,t}, n_{2,t}, \dots, n_{r,t})$, where $n_{i,t}$ is the number of tier-i students on the waitlist. We assume $\mathbf{n}_0 = \mathbf{0}$.

We use the vector $\mathbf{m}_t = (m_{1,t}, m_{2,t}, \dots, m_{r,t})$ to track matriculated students, where $m_{i,t}$ is the number of tier-i students who have already accepted an offer. We assume $\mathbf{m}_0 = \mathbf{0}$. The number of remaining slots is denoted by $c_t = C - \sum_{i=1}^{r} m_{i,t}$, where C is the maximum capacity of the school.

Admitted students are given a hard window of W periods to respond to an offer. The outstanding offers at time t are tracked via an $r \times W$ matrix $\mathbf{O}_t$, where the element $o_{i,t}^{w}$ represents the number of pending offers to tier-i students that have exactly $w \in \{1, 2, \dots, W\}$ periods of validity remaining (including the current period). We assume $\mathbf{O}_0 = \mathbf{0}$.

At any time t, the exact state of the system is described by the tuple $S_t = (\mathbf{m}_t, \mathbf{n}_t, \mathbf{O}_t)$.

2 Time-Dependent Transition Probabilities

To align the model with historical empirical data, all transition and arrival dynamics are generalized to be time-dependent.

- **Applicant Arrivals:** In each period t, at most one new applicant enters the system. Let $p_i(t)$ denote the probability that a student of tier i arrives at time t. To ensure a valid probability distribution, we require:

$$\sum_{i=1}^{r} p_i(t) \leq 1 \quad \forall t \in \{1, 2, \dots, T\}$$

  The probability that no applicant arrives during period t is given by $1 - \sum_{i=1}^{r} p_i(t)$.

- **Waitlist Dropouts:** A student of tier i remaining on the waitlist at time t independently drops out of the admissions process with probability $\lambda_i(t)$.
- **Outstanding Offer Responses:** Let $w \in \{1, 2, \ldots, W\}$ represent the number of periods remaining before an offer's decision deadline (including the current period). At time t, a tier-i student holding an offer with w periods remaining will:
  - **Accept** the offer with probability $\theta_i(t, w)$,
  - **Reject** the offer with probability $\mu_i(t, w)$,
  - **Remain undecided** with probability $1 - \theta_i(t, w) - \mu_i(t, w)$.

  Because students cannot remain undecided once the decision window expires, we enforce the boundary constraint:

  $$\theta_i(t, 1) + \mu_i(t, 1) = 1 \quad \forall i, \forall t$$

  For all periods with remaining validity ($w > 1$), the probability of remaining undecided is strictly non-negative, implying $\theta_i(t, w) + \mu_i(t, w) \leq 1$.

3 Timeline and Action Constraints

Let $S_t = (\mathbf{m}_t, \mathbf{n}_t, \mathbf{O}_t)$ denote the exact state of the admissions system at the beginning of period t. Within each period, events occur in the following chronological order:

1. **Arrival Phase:** A single applicant of tier i arrives with probability $p_i(t)$ and is added to the waitlist. We define the intermediate, post-arrival waitlist vector as $\mathbf{n}'_t$, where:

   $$n'_{j,t} = \begin{cases} n_{j,t} + 1 & \text{if a tier-}i\text{ student arrives and } j = i, \\ n_{j,t} & \text{otherwise.} \end{cases}$$

2. **Decision Phase:** The school observes the post-arrival state and selects an admission action vector $\mathbf{k}_t = (k_{1,t}, \ldots, k_{r,t})$, where $k_{i,t}$ is the number of offers extended to tier-i waitlist applicants. The chosen action must belong to the feasible action space $\mathcal{K}(S_t)$, defined by:

   $$\mathcal{K}(S_t) = \left\{ \mathbf{k}_t \in N_0^r \;\middle|\; k_{i,t} \leq n'_{i,t} \;\; \forall i, \text{ and } \sum_{i=1}^{r} k_{i,t} \leq c_t \right\}$$

   The state variables are immediately modified to their post-decision values:

   - Post-decision waitlist: $\tilde{n}_{i,t} = n'_{i,t} - k_{i,t}$
   - Post-decision offers at the maximum window: $\tilde{o}^W_{i,t} = o^W_{i,t} + k_{i,t}$ (for all shorter horizons $w < W$, the offer counts remain unaltered: $\tilde{o}^w_{i,t} = o^w_{i,t}$).

3. **Stochastic Response Phase:**

   - Remaining waitlist students ($\tilde{\mathbf{n}}_t$) independently choose to drop out of the process or remain.
   - Holders of outstanding offers ($\tilde{\mathbf{O}}_t$) execute their choices to accept, reject, or remain undecided.

4. **Clock Ticks:** Undecided offers age down by one period of validity, and the system clock increments from $t \to t+1$.

4 Stochastic Transition Dynamics

To simulate the state's transition from time t to $t+1$, we sample from Binomial and Multinomial distributions based on the post-decision variables.

4.1 Waitlist updates

Students on the post-decision waitlist independently drop out with probability $\lambda_i(t)$. The number of remaining students follows a Binomial distribution:

$$n_{i,t+1} \sim \text{Binomial}(\tilde{n}_{i,t},\, 1 - \lambda_i(t)) \quad \forall i \in \{1, \dots, r\}$$

4.2 Outstanding Offer Decisions

For each tier i and each remaining offer validity window $w \in \{1, \dots, W\}$, the pool of post-decision outstanding offers $o^w_{i,t}$ splits into three disjoint components. Let:

- $A^w_{i,t}$ be the number of students who accept the offer at time t,
- $R^w_{i,t}$ be the number of students who reject (or let expire) the offer at time t,
- $U^w_{i,t}$ be the number of students who remain undecided (do nothing) at time t.

These outcomes are jointly modeled using a Multinomial distribution:

$$\left(A^w_{i,t},\ R^w_{i,t},\ U^w_{i,t}\right) \sim \text{Multinomial}(\tilde{o}^w_{i,t},\ [\theta_i(t,w),\, \mu_i(t,w),\, 1 - \theta_i(t,w) - \mu_i(t,w)])$$

Recall that $\theta_i(t,1) + \mu_i(t,1) = 1$, so $U^1_{i,t} = 0$.

4.3 State Progression to $t+1$

Once the stochastic variables are sampled, the state components at the start of period $t+1$ are compiled.

The matriculation vector accumulates all new acceptances across outstanding cohorts:

$$m_{i,t+1} = m_{i,t} + \sum_{w=1}^{W} A_{i,t}^{w} \quad \forall i \in \{1, \ldots, r\}$$

The outstanding offer matrix updates by incrementing the undecided segments down by one period. For all $i \in \{1, \ldots, r\}$, the updated matrix elements are given by:

$$o_{i,t+1}^{w} = \begin{cases} U_{i,t}^{w+1} & \text{if } w < W \\ 0 & \text{if } w = W \end{cases}$$